JAY MCKNIGHT

30 Ballantine Lane, Lisburn BT27 5FG | Irish Citizen (EU Passport) - No Visa Required | 07718597886 | jaymcknight@icloud.com | https://main--jay-mcknight.netlify.app/

As an Irish citizen with full EU working rights, I am willing to relocate immediately and require no visa sponsorship.

Professional Summary
Full-Stack Software Engineer with a First Class Honors degree and proven commercial experience delivering robust, **cloud-integrated** solutions. Specializes in building resilient backend systems using **C# .NET Core and Azure Cloud Services (Functions, Service Bus)**, alongside modern frontends in **React** and **Next.js**. Demonstrated success in integrating complex enterprise systems (TMS/EDI), optimizing API performance, and championing Agile methodologies.

EXPERIENCE

Software Engineer (Full Stack) | Woodside Logistics | Belfast
2024 – Present

- Designed and maintained RESTful APIs that are integrated with the company's Transportation Management System.
- Architected and deployed a cloud-native EDI integration for a Global Fortune 500 partner, leveraging Azure Blob Storage and Event-Driven Azure Functions to automate high-volume data ingestion.
- Created responsive and interactive web interfaces using Next.js, React, and JavaScript.
- Built and deployed Docker containers onto Linux servers for in-house processes.
- Built and maintained relational SQL database tables, ensuring safe and efficient data storage.
- Connected and implemented SFTP connections with car manufacturers into windows worker services.
- Championed the implementation of agile methodologies within a small developer team.
- Successfully implemented EDI connections for multiple car manufacturers, ensuring real-time data exchange and compliance.
- Engineered robust EDI integrations using RabbitMQ, Polly (for resilience), and Dead Letter Queues to ensure 100% data retention during API transmission.
- Utilized GoLang for developing small-scale internal projects and utilities.

Junior Software Engineer (C# Developer) | Verint | Belfast
2022 – 2024

- Collaborated with senior developers to implement new features using a TDD approach.
- Integrated SonarQube to identify and mitigate security issues, reducing threats by 10%.
- Developed and optimized SQL stored procedures within SSMS.
- Performed end-to-end regression testing for major releases, including test automation.
- Conducted daily code reviews for team members' pull requests.

PROJECTS

SoberStride | Queen's University Belfast | Final year project (.NET Core Razor pages)

- Developed a support application for substance abuse issues with journaling, progress tracking, and resource recommendations using .NET Core Razor Pages, Bootstrap, and SQL Server.
- Created initial design **storyboards** using Figma.
- **Designed** and **implemented unit tests** to ensure high usability.
- Managed **version contro**l with GitHub and tracked progress with Jira.

NotiFly| Flight Price Monitoring Service | Personal Project | Serverless Cloud Architecture

- Architected a **serverless backgroun**d service using .NET 8 **Azure Functions** (Isolated Worker) to automate recurring flight price analysis for Southeast Asia routes.
- Integrated **third-party APIs** (Amadeus Flight Offers & Meta WhatsApp Business) to fetch real-time pricing and deliver instant **mobile alerts** via structured JSON payloads.
- Optimized API quota usage by implementing efficient scheduling strategies and an in-memory static cache for airline reference data, reducing external calls by over 90%.

QuietMail | Email Cleanup Web Application | Personal Project | In Progress...

- **Developed a full-stack email management platform** using **.NET** and **React**, featuring fault-tolerant batch orchestration that isolates sender-specific errors to maintain workflow stability.
- **Architected asynchronous processing pathways** integrated with a message queue system to handle high-volume background tasks and automated cleanup workflows efficiently.
- **Delivered a responsive frontend** enhanced with **SignalR WebSockets** for real-time scan progress updates and automated state sanitization for clean selection handling.

TECHNICAL SKILLS

- **Core Stack:** C#, .NET, ASP.NET Core, Web API, Entity Framework, SQL Server, React.
- **Frontend:** JavaScript (ES6+), Next.js, HTML5, Bootstrap.
- **DevOps & Tools:** Cloud & DevOps: Azure (Functions, Blob Storage, Service Bus), Docker, RabbitMQ, Git, Azure DevOps/Jenkins, SonarQube, Postman, Watchtower

EDUCATION

First Class Honors Bachelor of Science in Computing and Information Technologies | Queen's University Belfast
September 2020 – June 2024